

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1704

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

I MITHRA DEVI S - 192524203 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

S. Mithra Devi

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- II. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- III. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- IV. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$
- 2) P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$
- 3) P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: $\text{SoilDry} = \text{True}, \text{CropWheat} = \text{True}$

- (a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).
- (b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.
- (c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) Stench [1,2] $\rightarrow$ Wumpus is adjacent to cell [1,2]
- 2) Breeze [1,1] $\rightarrow$ Pit is adjacent to cell [1,1]
- 3) Glitter [x,y] $\rightarrow$ Gold is at [x,y]
- 4) $\neg$ Wumpus [x,y] $\wedge$ $\neg$ Pit[x,y] $\rightarrow$ Safe[x,y]

- (a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.
- (b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.
- (c) Evaluate whether the path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

CASE STUDY 1 – SMART MEDICAL DIAGNOSIS SYSTEM

Problem Understanding

The hospital uses a rule-based logical agent for preliminary diagnosis. The knowledge base contains rules connecting patient symptoms with possible diseases. Patient A has fever, cough, and breathlessness. The objective is to use Propositional Logic, Forward Chaining, and Backward Chaining to determine the possible diagnoses.

(a) Representation Using Propositional Logic

Step 1: Define Propositional Symbols

- F = Patient A has Fever.
- C = Patient A has Cough.
- R = Patient A has Rash.
- B = Patient A has Breathlessness.
- FL = Patient A has Possible Flu.
- M = Patient A has Possible Measles.
- P = Patient A has Possible Pneumonia.

Step 2: Represent the Rules

Rule 1: If a patient has Fever AND Cough, then Possible Flu.

$$F \wedge C \rightarrow FL$$

Rule 2: If a patient has Fever AND Rash, then Possible Measles.

$$F \wedge R \rightarrow M$$

Rule 3: If a patient has Cough AND Breathlessness, then Possible Pneumonia.

$$C \wedge B \rightarrow P$$

Step 3: Represent the Facts

$$F = \text{True}, C = \text{True}, B = \text{True}$$

$$KB = \{F, C, B, (F \wedge C \rightarrow FL), (F \wedge R \rightarrow M), (C \wedge B \rightarrow P)\}$$

(b) Forward Chaining

Forward Chaining starts with known facts and repeatedly applies rules whose conditions are satisfied.

Initial Facts: F, C, B (Fever = True, Cough = True, Breathlessness = True).

Inference Step 1: Rule 1: $F \wedge C \rightarrow FL$. Since Fever and Cough are true, $FL = \text{True}$. Derived fact: Patient A has Possible Flu.

Inference Step 2: Rule 2: $F \wedge R \rightarrow M$. Fever is true, but no Rash fact is available. Therefore, Rule 2 cannot fire and Measles cannot be derived.

Inference Step 3: Rule 3: $C \wedge B \rightarrow P$. Since Cough and Breathlessness are true, $P = \text{True}$. Derived fact: Patient A has Possible Pneumonia.

Step	Known Facts	Rule Fired	New Fact
0	F, C, B	None	Initial facts
1	F, C	Rule 1	FL
2	F, R	Rule 2 cannot fire	None
3	C, B	Rule 3	P

Final Conclusions: Possible Flu = True; Possible Pneumonia = True; Possible Measles = Not Derived.

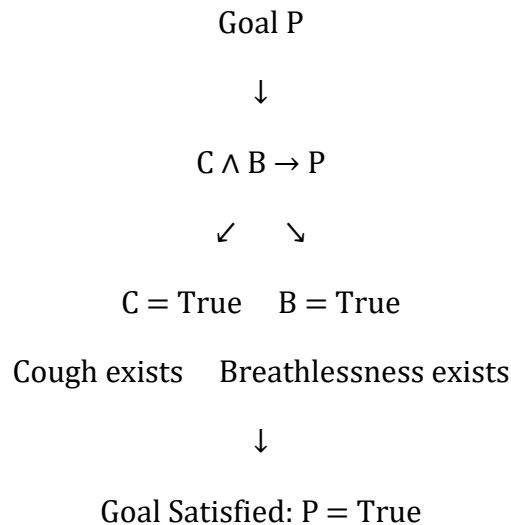
(c) Backward Chaining for Pneumonia

Goal: P, where P means Patient A has Possible Pneumonia.

Step 1: Find a rule producing P. Rule 3 is $C \wedge B \rightarrow P$. Therefore the goal P is reduced to $C \wedge B$.

Step 2: C is a known fact because Patient A has Cough.

Step 3: B is a known fact because Patient A has Breathlessness.



Conclusion: Both required conditions are known facts. Therefore, Backward Chaining confirms Possible Pneumonia for Patient A.

CASE STUDY 2 – AUTONOMOUS TRAFFIC MANAGEMENT AGENT

Problem Understanding

The traffic management system uses First-Order Logic to control traffic during emergencies. An ambulance is classified as an emergency vehicle. The system clears its path, provides a green signal, and allows vehicles behind it to proceed.

(a) Unification

Rule 1:

$$\forall x \text{ Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$$

Facts:

$$\text{Vehicle}(\text{Ambulance})$$

$$\text{EmergencyType}(\text{Ambulance})$$

Step 1: Match $\text{Vehicle}(x)$ with $\text{Vehicle}(\text{Ambulance})$. Therefore $x/\text{Ambulance}$.

Step 2: Match $\text{EmergencyType}(x)$ with $\text{EmergencyType}(\text{Ambulance})$. Again $x/\text{Ambulance}$.

$$\text{MGU: } \theta = \{x/\text{Ambulance}\}$$

$$\text{Vehicle}(\text{Ambulance}) \wedge \text{EmergencyType}(\text{Ambulance}) \rightarrow \text{ClearPath}(\text{Ambulance})$$

Therefore, $\text{ClearPath}(\text{Ambulance})$ is derived.

(b) Resolution Proof of Proceed(CarA)

Step 1: Convert Rules into CNF

$$\text{Rule 1: } \neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$$

$$\text{Rule 2a: } \neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$$

$$\text{Rule 2b: } \neg \text{ClearPath}(x) \vee \neg \text{RedSignal}(x)$$

$$\text{Rule 3: } \neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$$

Relevant facts: $\text{Vehicle}(\text{Ambulance})$, $\text{EmergencyType}(\text{Ambulance})$, $\text{Vehicle}(\text{CarA})$, $\text{Behind}(\text{CarA}, \text{Ambulance})$.

Step 2: From $\text{Vehicle}(\text{Ambulance})$ and $\text{EmergencyType}(\text{Ambulance})$, Rule 1 derives $\text{ClearPath}(\text{Ambulance})$.

Step 3: From $\text{ClearPath}(\text{Ambulance})$ and Rule 2a, derive $\text{GreenSignal}(\text{Ambulance})$.

Step 4: Substitute $x = \text{CarA}$ and $y = \text{Ambulance}$ in Rule 3. Using $\text{Vehicle}(\text{CarA})$, $\text{Behind}(\text{CarA}, \text{Ambulance})$, and $\text{GreenSignal}(\text{Ambulance})$, resolution derives $\text{Proceed}(\text{CarA})$.

$\therefore \text{Proceed}(\text{CarA})$

Conclusion: Resolution proves that CarA can proceed because it is behind the ambulance and the ambulance has a green signal.

(c) Handling Two Simultaneous Emergency Vehicles

The existing knowledge base can syntactically handle multiple vehicles because the rules are universally quantified. However, it does not contain rules for resolving conflicts when two emergency vehicles require the same road or intersection.

Additional rules may include:

$$\text{EmergencyVehicle}(x) \wedge \text{HigherPriority}(x,y) \rightarrow \text{GivePriority}(x,y)$$
$$\text{Emergency}(x) \wedge \text{Emergency}(y) \wedge \text{SameIntersection}(x,y) \rightarrow \text{ResolvePriority}(x,y)$$
$$\text{Conflict}(x,y) \rightarrow \neg \text{GreenSignal}(y)$$

Additional facts should specify current location, direction of travel, destination, priority level, intersection occupancy, and whether vehicles approach the same road segment.

Conclusion: The current KB is sufficient for independent emergency vehicles but insufficient for conflicting simultaneous emergencies. Priority, conflict-detection, location, and traffic-state rules are required.

CASE STUDY 3 – AGRICULTURAL EXPERT REASONING SYSTEM

Problem Understanding

The agricultural advisory system determines whether irrigation is needed and whether the drip irrigation method should be applied to wheat crops.

(a) CNF Conversion

Let S = SoilDry, I = IrrigationNeeded, W = CropWheat, D = ApplyDripMethod, and A = CropAtRisk.

P1

$$S \rightarrow I$$

$$\neg S \vee I$$

P2

$$I \wedge W \rightarrow D$$

$$\neg(I \wedge W) \vee D$$

$$\neg I \vee \neg W \vee D$$

P3

$$\neg D \wedge W \rightarrow A$$

$$\neg(D \wedge W) \vee A$$

$$\neg D \vee \neg W \vee A$$

Facts

$$S$$

$$W$$

Complete CNF Knowledge Base

$$C1: \neg S \vee I$$

$$C2: \neg I \vee \neg W \vee D$$

$$C3: \neg D \vee \neg W \vee A$$

$$C4: S$$

$$C5: W$$

(b) Resolution Refutation for ApplyDripMethod

Goal: D = ApplyDripMethod.

Negate the goal and add $\neg D$ to the KB.

C6: $\neg D$

Resolve C4: S with C1: $\neg S \vee I$.

C7: I

Resolve C7: I with C2: $\neg I \vee \neg W \vee D$.

C8: $\neg W \vee D$

Resolve C8 with C5: W.

C9: D

Finally, resolve C9: D with C6: $\neg D$.

□

The empty clause indicates contradiction. Therefore KB entails ApplyDripMethod.

KB $\models$ ApplyDripMethod

Conclusion: The agricultural reasoning system logically concludes that the drip irrigation method should be applied.

(c) Removing SoilDry

If the fact SoilDry is removed, the rule $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$ remains, but its antecedent is no longer established.

Step 1: SoilDry becomes unknown.

Step 2: IrrigationNeeded cannot be derived.

Step 3: Since IrrigationNeeded is unavailable, ApplyDripMethod cannot be derived from P2, even though CropWheat is true.

Step 4: CropAtRisk also cannot be derived because P3 requires explicit $\neg \text{ApplyDripMethod}$. Failure to prove ApplyDripMethod does not mean $\neg \text{ApplyDripMethod}$.

Final conclusion: IrrigationNeeded is not derivable; ApplyDripMethod is not derivable; $\neg \text{ApplyDripMethod}$ is not derivable; CropAtRisk does not logically follow.

CASE STUDY 4 – WUMPUS WORLD KNOWLEDGE AGENT

Problem Understanding

The Wumpus World agent receives three percepts: Stench at [1,2], Breeze at [1,1], and Glitter at [2,2]. The agent uses logical inference to determine possible danger locations and whether the proposed path is safe.

(a) Belief State and Modus Ponens

Initial percepts:

Stench[1,2]

Breeze[1,1]

Glitter[2,2]

Rule 1: Stench[1,2] $\rightarrow$ WumpusAdjacent[1,2]. By Modus Ponens, WumpusAdjacent[1,2].

Rule 2: Breeze[1,1] $\rightarrow$ PitAdjacent[1,1]. By Modus Ponens, PitAdjacent[1,1].

Rule 3: Glitter[x,y] $\rightarrow$ Gold[x,y]. For x = 2 and y = 2, Gold[2,2] is derived.

Rule 4 requires explicit $\neg$ Wumpus[x,y] and $\neg$ Pit[x,y]. No such negative facts are supplied, so Safe[x,y] cannot be derived.

Belief State

KB = {Stench[1,2], Breeze[1,1], Glitter[2,2], WumpusAdjacent[1,2], PitAdjacent[1,1], Gold[2,2]}

(b) Forward Chaining – Possible Wumpus and Pit Locations

Step 1: Stench[1,2] implies that a Wumpus is adjacent to [1,2]. Under the standard grid assumption, possible neighboring cells include [1,1], [1,3], and [2,2] (subject to the actual world boundaries).

Step 2: Breeze[1,1] implies a pit is adjacent to [1,1]. In a standard grid, possible pit locations include [1,2] and [2,1].

Step 3: Glitter[2,2] implies Gold[2,2].

Conclusion: The exact Wumpus location cannot be identified from the given information. The possible pit locations include [1,2] and [2,1].

(c) Safety of the Path

Proposed path: [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2].

Cell [1,1]: Breeze[1,1] indicates that a pit is adjacent. The cell itself is not proven to contain a pit, but its neighboring cells require caution.

Cell [1,2]: It is a possible pit location because of Breeze[1,1]. Stench[1,2] also indicates a Wumpus is adjacent to [1,2]. Therefore Safe[1,2] cannot be established.

Cell [2,2]: Glitter[2,2] establishes Gold[2,2], but glitter does not establish safety. The KB does not explicitly establish $\neg$ Wumpus[2,2] and $\neg$ Pit[2,2].

Cell	Knowledge	Safety
[1,1]	Breeze indicates nearby pit	Not fully proven safe
[1,2]	Possible pit location; stench present	Unsafe / cannot be proven safe
[2,2]	Gold present	Safety not proven

Final Decision: The path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ cannot be logically classified as safe using the current knowledge base. In particular, [1,2] is a possible pit location. The agent should not confidently select this path unless additional percepts or rules establish $\neg$ Pit[1,2] and $\neg$ Wumpus[1,2], or otherwise provide sufficient safety evidence.

Final conclusion: The proposed path is NOT proven safe.

OVERALL CONCLUSION

The four case studies demonstrate how Artificial Intelligence agents use different knowledge representation and reasoning techniques to make logical decisions.

Case Study 1 – Medical Diagnosis: Propositional Logic, Forward Chaining, and Backward Chaining were used to identify possible diseases. Patient A was logically identified as having possible Flu and possible Pneumonia.

Case Study 2 – Traffic Management: First-Order Logic and Unification were used to match variables with specific vehicles. Resolution demonstrated that CarA can proceed because it is behind the ambulance and the ambulance receives a green signal.

Case Study 3 – Agricultural Advisory System: The rules were converted into CNF and Resolution Refutation was used to prove ApplyDripMethod. When SoilDry was removed, the system could no longer derive IrrigationNeeded or ApplyDripMethod. CropAtRisk also could not be concluded because failure to prove ApplyDripMethod does not imply its negation.

Case Study 4 – Wumpus World: The agent used percepts and Modus Ponens to derive information about the Wumpus, pit, and gold. The analysis showed that the proposed path cannot be proven safe because [1,2] is a possible pit location.

Overall, these case studies demonstrate the importance of knowledge representation, inference, unification, forward chaining, backward chaining, and resolution in building intelligent agents capable of logical reasoning and decision making.